front matter

Preface

I’m fortunate to have been a computer scientist during an era of phenomenal advancement in computing, from integrated chips to the internet. I have designed and developed a wide range of systems, from a dot-matrix printer driver to algorithms for fault tolerance in distributed systems. I have programmed in a variety of high-level languages, from PL/1 to Python. All these years, I have also been an educator, teaching courses on the leading edge of technology, from grid computing to data science. And of course, my current passion and fascination is blockchain technology.

I first heard about Bitcoin around 2013, but ignored it as being yet another attempt at cryptocurrency. In 2016, I went back to explore Bitcoin for its underlying technology: the blockchain. I searched for more information on blockchain, of course, but could not find much. At a local meetup in Buffalo on a cold January night in 2016, one of the speakers showed a few YouTube videos on the magic of the blockchain’s distributed ledger. That was my “Aha!” moment. I was amazed. I went on to read the Bitcoin white paper; then I dabbled with open source blockchain code with Eris and (later) Monax. In the summer of 2017, I taught blockchain in a course on Emerging Technologies. This